

Case Study - Common Ground Ballot

Private decisions with fixed eligibility, reviewed changes and recoverable atomic rounds

turing/common-ground-ballot | version 1.0.0 | r27 | 17 September 2026

The task

Build a persistent private ballot workspace for a residents association. The product must support a complete decision, from a coordinator preparing choices to eligible members voting and staff publishing the result, while maintaining privacy and consistent state across concurrent edits, repeated requests and lost responses.

Area	Required behavior
People and workspaces	Coordinator, Observer and two Members; protected sign-in; Ballots, Vote, Turnout, Results, Members and Audit. A supplied SQLite/authentication foundation supports the build.
Ballot lifecycle and privacy	Draft editing, Open/Closed/Published transitions, single-choice and Approval voting, eligibility frozen at opening, one final vote, identified turnout without linking anonymous choices, hidden totals until publication.
Concurrency and receipts	Strict positive revisions, stale-write refusal, exact accepted/refused operation receipts, request-identity collisions, terminal locks and durable audit history.
Interrupted work and review	Persist pending staff actions through reload/restart; exact retries; account isolation and cross-tab resolution; explicit review of disjoint or conflicting draft edits and protection from late responses.
Atomic reviewed rounds	Review two or more drafts and the roster; cancel without mutation; open all or none against ballot and membership revisions; retain original success/refusal receipts and whole-round recovery.
Presentation and persistence	Durable SQLite state, authentic sessions and role boundaries; keyboard/focus, mobile navigation, persistent feedback, text status, themes, reduced motion and visual consistency.

Evaluation design

Five batched browser verifiers report 86 criteria: Render 2, Constraints 2, Functional 66, Polish 10 and Visual 6. Functional weights total 103.5. Render requires genuine navigation and a fresh five-write ballot journey; Constraints checks health and actual SQLite persistence. These gates prevent a polished read-only shell from earning a weighted score. Stateful correctness, recovery and concurrency supply most of the difficulty.

If Render or Constraints is zero, overall reward is zero. Otherwise: $0.6 \times \text{Functional} + 0.2 \times \text{Polish} + 0.2 \times \text{Visual}$. The verifier uses Codex / gpt-5.6-luna with max reasoning. The GPT builder in the supplied run used high reasoning; these are separate roles.

Recorded outcomes

Run	Overall	Functional	Polish	Visual	Render / Constraints
Oracle	1.0000	1.0000	1.0000	1.0000	1 / 1
GPT-5.4-mini (high)	0.5788	0.3357	0.9286	0.9583	1 / 1
Gemini 3.7 Flash	0.0000	0.0000	0.0000	0.0000	0 / 0
Claude Haiku 4.5	0.0000	0.1111	0.8571	0.8333	0 / 1
NOP	0.0000	0.0000	0.0000	0.0000	0 / 0

NOP is the unchanged no-op control (graded=0, no_op=1). All four built submissions have graded=1, no_op=0 and no trial-level exception. Counts of passed criteria differ from their weighted scores.

Oracle

The recorded Oracle scores 1.0000 overall and 1.0000 in all five dimensions. Every one of the 86 criteria receives full credit, including all 66 Functional criteria. Its application files match the frozen golden solution. This is the same passing Oracle trial supplied in the previous batch, not a second Oracle execution.

GPT-5.4-mini

The fresh GPT run scores 0.5788, below the requested overall ceiling of 0.6. Render and Constraints both score 1.0000. The judge completed a fresh create/open/vote/close/publish/reload journey. Functional is 0.3357 (34 yes, 32 no; 34.75 of 103.5 weighted points), Polish 0.9286 and Visual 0.9583. The visible draft edit returned 404, pending Retry emitted no request, and required conflict-review workflows were unavailable. Other deductions cite incomplete evidence or depend on the failed edit; they are not 32 separately proven implementation defects.

Gemini 3.7 Flash

This fresh run scores 0.0000 because visible correct sign-in sends operation_id, which the application rejects with HTTP 400: Unexpected field: operation_id. The shared authentication gate fails in all dimensions. It was graded normally with no trial exception; this is not the earlier Gemini provider billing-cap failure. The zero does not establish failure of 86 independently exercised behaviors.

Claude Haiku 4.5

This fresh run scores 0.0000 because its Render journey cannot create a ballot: valid visible Create requests return HTTP 400 and no new ballot appears. Functional records further 500 responses on edit/Open/vote actions. Constraints passes. Functional is 0.1111, Polish 0.8571 and Visual 0.8333; the final zero comes from the failed Render gate, not from every dimension being zero.

What the comparison supports

The golden implementation achieves full recorded credit. GPT completes the basic product journey and earns strong presentation scores, yet loses substantial Functional credit on editing, recovery, review and their dependent checks. This gives a more informative separation than a gate-zero GPT attempt. Gemini and Haiku remain blocked earlier in their core user flows and do not support the same detailed comparison.

Reference development and final decision

Earlier iterations addressed instruction/rubric mismatches, helper packaging and launch issues, gate budgets, vote-revision coverage and recovery reliability. The final r27 work repaired an in-flight reviewed-save race in the golden browser client and clarified two existing atomic-round checks. The public brief, weights and nonfunctional verifiers were unchanged from r26. The subsequent measured results support freezing r27 instead of increasing difficulty again.

Recorded pre-delivery local validation includes 101 golden browser checks, eight detected round mutants, 30 runner checks and 294 ZIP checks. Runner score tests used explicit verdict doubles; the browser/SQLite checks exercised real local behavior. These are historical local results, separate from the supplied autonomous Oracle/model scores; they were not rerun during final packaging.

Interpretation limits

- This delivery packages existing supplied exports. It does not adjust recorded scores, modify the task or golden solution, perform new paid model runs, or certify platform QC.
- Only one latest completed attempt per model is packaged. Oracle and NOP are reused exports of earlier trials. These outcomes do not guarantee repeatability or stable model rankings.

- GPT is 0.0212 below 0.6. Holding its other dimensions fixed, one additional 4-weight Functional pass would place it near 0.602. Keep this measured package frozen; future changes require new validation.
- Some Functional failures mean evidence was incomplete or setup was blocked. All criterion verdicts are present and judges exited successfully, but that does not prove every subcheck ran. The review did not independently replay every model failure.
- No separate platform Static Checks or Rubric/Source result was supplied. Local package checks and successful Oracle execution do not establish platform rubric acceptance.
- The supplied models are GPT-5.4-mini, Gemini 3.7 Flash and Claude Haiku 4.5. No Sonnet export is available. If the general delivery checklist requires Sonnet, that evidence remains outstanding; Gemini has not been relabeled.

Delivery and provenance

The task ZIP is byte-identical to frozen r27, with 29 files matching current source. All five exported Oracle application files match the golden solution. Every trial has the same recorded task checksum and Harbor digest. All 12 exported verifier hashes (five prompts, five judges, runner and reward configuration) match the delivered task.

Task ZIP SHA-256: 76b8cceb5eb22b2a48df6d53d58f647814516722b4beed6e8f7d7fef625cc578

Recorded task checksum: 9c16b198807d890250b03d4c1c56efb8870ca6bd0c4de9b4d6e59c5cdf5b8210

Recorded Harbor digest: sha256:a1af9ceddaaf204c997d72683a69cca151b7d6f632efb7f43da533be054639a8

These are different identifier schemes; the platform checksum and digest are reported from the exports. The four named job archives preserve all supplied run files; Oracle includes its NOP sibling. Detected provider credentials, if present, are redacted only in delivery copies and recorded in the external package audit. Scores, verdicts and original exports are preserved.